

VIVEK KUMAWAT

AI/ML Engineer | Backend Systems | Agentic Development

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SUMMARY

AI/ML Engineer and Computer Science undergraduate focused on backend systems, applied deep learning, and agentic development workflows. Comfortable owning a project end-to-end, from architecture and data pipelines to production-ready deployment.

EDUCATION

B.Tech in Computer Science & Engineering (AI & ML) — Jain (Deemed-to-be University), Bengaluru

2024 – 2028

SGPA: 9.29/10 (Semester 4)

PROJECTS

Ripple — Global Intelligence & Predictive Causal-Chain Analysis Platform

- Modeled global events as an interconnected causal graph to trace direct and downstream impacts across economies, trade, and geopolitics.
- Engineered evidence-grounded AI analysis with **confidence scoring** and **contradiction handling** using **Gemini** and **Groq**.
- Optimized visualization for **1,000+ events** with an async **FastAPI/PostgreSQL** backend integrated with a **React/Three.js** interface.

Winter — Local-First AI Desktop Companion (In Development)

- Designing a local-first desktop AI companion using **Tauri 2** with a deterministic core and contextual memory via **SQLite** and embeddings.
- Architecting a plugin-based system for AI agents and developer tools with local/BYOK model support and event-driven architecture.

Vaani — Voice Authenticity Detection

- Achieved **90% test accuracy** in human vs. AI-generated speech classification with a fusion model combining **Wav2Vec2** embeddings and acoustic signal features.
- Improved system reliability by implementing entropy-based uncertainty scoring to flag low-confidence predictions.
- Built the full-stack inference pipeline using **FastAPI** and **React/TypeScript** for real-time audio analysis.

SkySense AI — Full-Stack AI Weather Intelligence Platform

- Developed a weather intelligence platform featuring a conversational assistant (Nimbus) powered by **Groq** and **scikit-learn**.
- Implemented ML inference, caching, and API failover mechanisms around a **FastAPI** backend to ensure high availability and **<1s latency**.

EXPERIENCE

AI/ML Intern — Harzio

Jun 2026 – Jul 2026

- Delivered **five AI/ML projects**, including a student-score regression model ($R^2 = 0.919$) and a Decision Tree fraud classifier (**99.9% accuracy**).
- Conducted deep exploratory data analysis and model evaluation across multiple classification and regression tasks.

RESEARCH

Real-Time Water Quality Monitoring Using IoT and Anomaly Detection

Interdisciplinary Research Project, JAIN University — Co-authored, Unpublished Prototype

- Achieved **F1 = 0.936** and **ROC-AUC = 0.926** on contamination detection with an AI pipeline using **Isolation Forest** and **Kalman filtering**.
- Owned the backend architecture using **MQTT** telemetry, **FastAPI**, and **InfluxDB** for real-time monitoring and **Grafana** visualization.

ACHIEVEMENTS

- First Runner-Up**, TechSprint — a hackathon organized by GDG on Campus, FET JU — built AeroMon, an app that finds the least-polluted walking route using real-time air quality data.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI/ML: PyTorch, Scikit-learn, Hugging Face Transformers, LLMs, NLP, Deep Learning

Backend & Data: FastAPI, PostgreSQL, REST APIs, AWS, GCP, Git/GitHub

Agentic Workflow: Google Antigravity, OpenCode, Codex, agents.md, SKILL.md, MCP Servers